WELLSENSE · REGULATORY REGISTER

WellSense regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade); drinking water rows reuse the Water Security focus-area research (StillStack, LumaFlow)

WellSense regulatory and standards register

Which standards and regulations would apply to WellSense, where, and what is needed. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. The first deployment region is India with a participatory groundwater partner (WLS-DEC-001), so Indian rows lead; international rows cover the open design wherever it is built. No certification is pursued while the portfolio is capped at TRL 3. Drinking water rows reuse the Water Security focus-area research cited in StillStack (SSK-REG-001) and LumaFlow (LMF-REG-001). Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	NSF/ANSI/CAN 61, drinking water system components, health effects	United States and Canada (often accepted as evidence elsewhere)	Contaminants that leach from materials in contact with drinking water	Every wetted part: probe body, vented cable jacket, PVC tube and end cap, HDPE seal plate, EPDM gasket and wraps (PR-009, STM S24)	Design lead with suppliers	To assess: no certificate in hand for the low-cost parts (PR-009 not met)
G2	IS 10500:2012, drinking water specification (second revision), Bureau of Indian Standards	India	Acceptable and permissible limits for drinking water quality	Context: WellSense adds nothing to the water; its wetted parts must not move any parameter (G1); disinfection after installation (G5)	Partner and well owner	Applies as guidance
G3	WHO Guidelines for Drinking-water Quality, 4th edition incorporating the first and second addenda (2022)	International (guidance)	Microbial and chemical guidance, including protection of groundwater sources and sanitary inspection of wells	Keeping the wellhead sealed (PR-009) and disinfecting anything lowered into a supply well	Design lead; partner	Applies as guidance
G4	Directive (EU) 2020/2184, Article 11 (materials in contact with water for human consumption)	European Union	Hygiene requirements for materials and products in contact with drinking water	Only if WellSense is supplied for EU drinking water wells	Whoever places it on the market	Does not apply to the open design; certification needed later for any EU supply
G5	AWWA C654-13, disinfection of wells	United States (practice reference)	Disinfecting wells and equipment put into them	Disinfection of the probe, cable, tube and seal before installation (STM S25, build plan safety stops)	Installer	Applies as practice reference; follow the local water authority where it differs
G6	CGWA Guidelines to regulate and control ground water extraction in India, notification S.O. 3289(E), 24 September 2020	India (pan-India where states do not regulate groundwater themselves)	No objection certificates for groundwater abstraction; piezometers for users drawing more than 10 m³/day,	Market context, not a requirement on community wells: shows the regulator already expects digital water level	Project lead	To assess

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			with a digital water level recorder from 51 m³/day and telemetry above 500 m³/day (Table 14.1)	recorders; whether a WellSense unit could serve as a compliant recorder is to assess with CGWA		
G7	ISO 4373:2022, hydrometry, water level measuring devices	International	Performance classes and test methods for water level measuring devices	A route to state WellSense's accuracy class at TRL 4 (PR-004, PR-005)	Design lead with a test laboratory	To assess (TRL 4 or later)
G8	USGS Techniques and Methods 1-A1 (Cunningham and Schalk, 2011), groundwater technical procedures	United States (method reference)	Measuring well water levels with a steel tape, an electric tape and a submersible pressure transducer	The method WellSense follows: transducer for the record, tape for the datum and checks (PR-004, PR-005)	Design lead	Applies as method reference
G9	IEC 60529, degrees of protection by enclosures (IP code)	International	Ingress protection ratings	Probe IP68, above-ground parts IP65 or better (PR-015); met on datasheets for bought parts	Design lead	Met by design (datasheets, to confirm at purchase)
G10	IEC 61140, protection against electric shock	International	Extra-low voltage bands and protective measures	WellSense runs at 24 V DC or less (PR-018, STM S27); the pump's mains supply is separate and isolated before work (build plan safety stops)	Electronics lead	Met by design
G11	Wireless Planning and Coordination Wing (WPC), Department of Telecommunications: licence-exempt use of 865 to 867 MHz and equipment type approval (ETA) for radio devices	India	Radio use and approval of the LoRaWAN radio in the FieldNode core	Carried by the FieldNode core on the IN865 band (WLS-DEC-001); WellSense adds no radio	FieldNode project	To assess (FieldNode repo)
G12	Electromagnetic compatibility rules for the product's market (for example the EU EMC	European Union; United States	Emissions and immunity of the boost converter and the logger	Only for a product placed on those markets	Electronics lead	To assess (product stage)

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	Directive 2014/30/EU or US FCC Part 15)					
G13	Local well construction and wellhead protection rules (state groundwater departments in India; state rules in the United States)	Local	Sealing, sanitary protection and who may open a well	Fitting the split seal plate and access tube to an existing well (PR-009, PR-010)	Partner and installer	To assess per site

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- Standards Council of Canada. "NSF/ANSI/CAN 61 - 2023." Standards database. https://scc-ccn.ca/standardsdb/standards/4032142 (checked 2026-10-24, StillStack SSK-REG-001).
- Bureau of Indian Standards. *IS 10500:2012, Drinking Water: Specification (Second Revision).* https://www.bis.gov.in/other/DrinWatIS10500.pdf (listed 2026-10-25).
- World Health Organization. *Guidelines for Drinking-Water Quality: Fourth Edition Incorporating the First and Second Addenda.* Geneva: WHO, 2022. https://www.who.int/publications/i/item/9789240045064 (checked 2026-10-24, StillStack SSK-REG-001).
- European Union. Directive (EU) 2020/2184 on the Quality of Water Intended for Human Consumption, Article 11. https://www.legislation.gov.uk/eudr/2020/2184/article/11 (checked 2026-10-24, StillStack SSK-REG-001).
- American Water Works Association. *AWWA C654-13, Disinfection of Wells.* https://store.awwa.org/AWWA-C654-13-Disinfection-of-Wells-PDF (listed 2026-10-25).
- Central Ground Water Authority, Ministry of Jal Shakti. *Guidelines to Regulate and Control Ground Water Extraction in India*, notification S.O. 3289(E), 24 September 2020, section 14 and Table 14.1. Copy hosted by the Maharashtra Pollution Control Board: https://www.mpcb.gov.in/sites/default/files/water-quality/standards-protocols/Ground_Water_NewGuidelinesNotifiedeng24092021102020.pdf (checked 2026-10-25).
- International Organization for Standardization. *ISO 4373:2022, Hydrometry: Water Level Measuring Devices.* AFNOR listing: https://www.boutique.afnor.org/en-gb/standard/iso-43732022/hydrometry-water-level-measuring-devices/xs139268/322212 (listed 2026-10-25).
- Cunningham, William L., and Charles W. Schalk. *Groundwater Technical Procedures of the U.S. Geological Survey.* Techniques and Methods 1-A1. Reston, VA: USGS, 2011. https://pubs.usgs.gov/tm/1a1/.
- Rows G9 to G13 are listed from the standards' and authorities' own titles; their detailed applicability is "to assess" and needs a test laboratory, the FieldNode project or a regulatory consultant. The LoRa band in India is described at https://www.everythingrf.com/community/lora-frequency-band-in-india (listed 2026-10-25); the WPC notification itself is to be cited by the FieldNode project.